Arman Akbari

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Professional Summary

PhD researcher in Efficient Physical AI and robotics, developing compression, distillation, and acceleration techniques that make World Action Models (WAMs) and Vision-Language-Action models (VLAs) fast and compact enough for real-time, on-robot deployment. Hands-on experience deploying these models on humanoids (Unitree G1, AgiBot) and robotic arms (UR3). Also working on privacy-preserving machine learning through the lens of efficiency.

Education

Northeastern University — Boston, MA
PhD in Computer Engineering | GPA: 3.9/4.00 — *Sep 2024 – Present*
- Advisors: Prof. Yanzhi Wang, Prof. Xuan Zhang

Northeastern University — Boston, MA
M.S. in Computer Engineering | GPA: 3.9/4.00 — *Sep 2024 – Aug 2026*

University of Tehran — Tehran, Iran
B.Sc. in Computer Science | GPA: 3.74/4.00 (1st rank) — *2020 – 2024*

Professional Experience

RoboForce — Milpitas, CA
AI Resident — *Aug 2026 – Present*
- Efficient World Action Models (WAMs) and VLAs for real-world robot deployment.

Northeastern University — Boston, MA
Graduate Research Assistant — *Sep 2024 – Present*
- World Action Models (WAMs), VLAs, Efficient AI, Multimodal learning, and trustworthy/privacy-preserving AI.

Singapore University of Technology and Design (SUTD) — Remote
Research Intern — *Jan 2023 – Apr 2023*
- Implemented Denoising Diffusion Probabilistic Models (DDPM), and evaluating under data-constrained regimes.

TaarLab: Human and Robot Interaction Laboratory — Tehran, Iran
Research Assistant — *Jun 2022 – Apr 2023*
- geometry-based algorithm for two-finger robotic grasping.

Selected Publications *('*' denotes equal contribution)*

- **Flash-WAM: Modality-Aware Distillation for World Action Models**
  Arman Akbari, et al. • *Under Review*
- **ActQuant: Sub-4-bit Action-Guided Quantization for Vision-Language-Action Models**
  Arash Akbari, Arman Akbari, et al. • *NeurIPS, 2026*
- **Palette: A Modular, Controllable, and Efficient Framework for On-demand Authorized Safety Alignment Relaxation in LLMs**
  Qitao Tan, Xiaoying Song, Arman Akbari, et al. • *NeurIPS, 2026*
- **CircuitSense: A Hierarchical Circuit System Benchmark Bridging Visual Comprehension and Symbolic Reasoning in Engineering Design Process**
  Arman Akbari, et al. • *ICLR, 2026*
- **Vote: Vision-Language-Action Optimization with Trajectory Ensemble Voting**
  Juyi Lin, Amir Taherin, Arash Akbari, Arman Akbari, et al. • *ICANN, 2026*
- **Holocode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training**
  Ruofan Xing, Arman Akbari, et al. • *DAC, 2026*

Honors and Awards

Ranked 1st in Computer Science Undergraduate Program (highest GPA, cohort of 42) — 2024

Ranked 355th (top 0.15%) in National University Entrance Exam (Konkur), Iran — 2020

Best Technical Team, RoboCup Asia-Pacific (RCJ Soccer Open League) — 2018

Skills

Languages & Frameworks: Python (PyTorch, TensorFlow, NumPy, Pandas), C++, Git, Linux, LaTeX
Technical Expertise: WAMs, VLAs, LLMs, Diffusion Language Models, MLLMs, VLMs, Computer Vision
Spoken Languages: English (IELTS: 7.0), Persian (Native)